Code-Layout, Readability and Reusability

Date	11 july 2026
Team ID	XXXXXX
Project Name	EV Data Analytics & Visualization Dashboard
Maximum Marks	5 Marks

Code-Layout, Readability and Reusability:

This document evaluates the quality of the codebase in terms of its structure, readability, and reusability. Well-organized code with proper naming conventions, comments, and modular design ensures that the project is maintainable and can be extended or reused in future development.

Code Layout Checklist:

S.No	Code Quality Parameter	Description	Followed (Yes / No / Partial)	Remarks
1	Consistent Indentation	Uniform spacing/tabs used throughout the code	Yes	Uniform 4-space indentation is maintained throughout the project.
2	Proper File Structure	Files and folders are logically organized	Yes	Frontend, backend, database, Tableau dashboards, and utility files are organized into separate folders.
3	Meaningful Variable Name	Variables reflect their purpose clearly	Yes	Variables such as charging_ data, battery_health, vehicle_range, and energy_consumption clearly describe their purpose.
4	Function / Method Names	Functions are descriptively named	Yes	Functions like load_dataset(), validate_data(), connect_database(), and generate_dashboard() are descriptive and easy to understand.
5	Code Comments	Inline and block comments explain logic	Yes	Important functions and data processing logic include meaningful comments.
6	Modular Design	Code is split into reusable functions/modules	Yes	The application is divided into reusable modules for data preprocessing, database connectivity, Tableau integration, and Flask web services.
7	No Redundant Code	Duplicate or unused code is removed	Yes	Reusable utility functions and common modules minimize code duplication.
8	Error Handling	Exceptions and errors are handled gracefully	Yes	Database connectivity, file uploads, and data validation include proper exception handling to improve reliability.

Reusable Components / Modules:

S.No	Component / Module Name	Language / Technolo	yWhere Reused	Reusability Level (High / Medium / Low)
1	Input Validation Module	Python (Flask)	Data Import & Preprocessing	High
2	Weather Service Module	Python, MySQL Connector	Flask Backend,Tableau Integration	High
3	Random Forest Prediction Module	Python (Pandas, NumPy)	Dashboard Analytics	High
4	File Upload Component	Tableau Desktop / Tableau Public	Charging Dashboard, Battery Dashboard, EV Comparison Dashboard	Medium
5	Recommendation Dashboard	Flask (Python)	Dashboard Embedding & Report Access	High

Overall Code Quality Assessment:

Aspect	Rating (1-5)	Comments
Code Layout & Structure	5	The project follows a clean and organized folder structure with separate frontend, backend, database, Tableau, and utility modules.
Readability	5	Descriptive variable names, consistent formatting, and clear function names make the code easy to understand and maintain.
Reusability	5	Common functionalities such as data validation, preprocessing, database connectivity, and dashboard integration are implemented as reusable modules.
Documentation / Comments	5	Important functions and modules are documented with comments, improving maintainability and future enhancements.
Overall Score	5.0	The **EV Data Analytics & Visualization Dashboard** follows software engineering best practices with modular design, reusable components, proper documentation, and scalable architecture, ensuring maintainability, readability, and future extensibility.